



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech., First Year

Minor 2 Examination, April-2026

Course Name: Basics of Mechanical and Civil Engineering

Date: 25th April 2026

Duration: 50 Minutes

Course Code: ME150

Time: 1:30 P.M. - 2:20 P.M.

Max. Marks: 30

(ALL QUESTIONS ARE COMPULSORY)

- Q.1.** A steel rod of 2 cm diameter is enclosed centrally in a hollow copper tube of external diameter 4 cm and internal diameter of 3.5 cm. The composite bar is then subjected to an axial pull of 50000 N. If the length of each bar is equal to 20 cm, determine: (i) the stress in the rod and tube, and (ii) load carried by each bar. Take E for steel = 2×10^5 N/mm² and for copper = 1×10^5 N/mm² (6 Marks)
- Q.2.** With the help of neat sketch explain working mechanism of ultrasonic machining process. (6 Marks)
- Q.3.** Two parallel shafts, connected by a crossed belt, are provided with pulleys 480 mm and 640 mm in diameters. The distance between the centre lines of the shafts is 3 m. Find the length of the (a) open and (b) closed belt drive. (6 Marks)
- Q.4.** Explain with neat sketch working process of electric arc welding. Also write its advantage, disadvantages and applications. (6 Marks)
- Q.5.** A shaft runs at 80 rpm and drives another shaft at 150 ~ rpm through belt drive. The ~ diameter of the driving pulley is 600 mm. Determine the diameter of the driven pulley in the following cases: (i) Neglecting belt thickness (ii) Taking belt thickness as 5 mm (iii) Assuming for case (ii) a total slip of 4% (iv) Assuming/or case (ii) a slip of 2% on each pulley (6 Marks)



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech. IV Year

Mid Semester Examinations, Septemebr-2025

Roll No	2	5	C	S	E	1	0	4	9
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Course Name: Basics of Mechanical and Civil Engineering

Course Code: ME150

Date: 28/11/2025

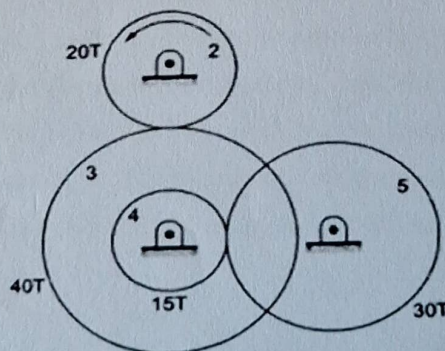
Time: 1:30 PM- 2:50 PM

Duration: 1hr. 20 min

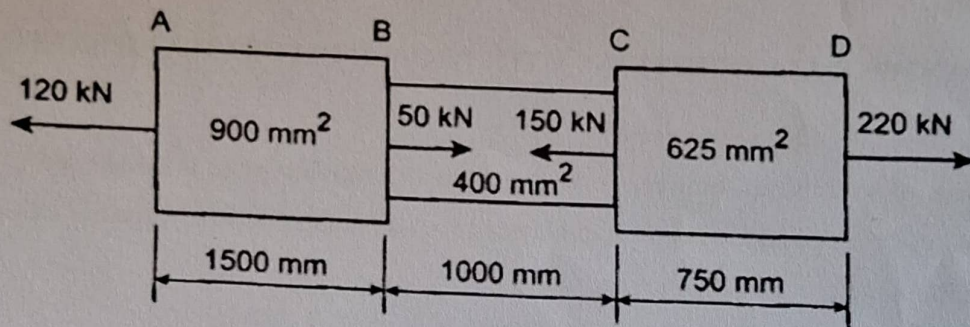
Max. Marks: 40

ANSWER ALL QUESTIONS WITH PROPER DIAGRAM AND SKETCH

1. Discuss the classification of Fluid? [2]
2. The rate of shear strain between a fixed plate and a moving plate is 24000 sec^{-1} . The viscosity of the fluid between the two plates is 5×10^{-4} poise. Find the force per unit area to maintain the speed of the moving plate. [5]
3. Discuss the module and circular pitch of gear? [2]
4. Find the length of Open belt drive? [3]
5. An open belt running over two pulleys 240 mm and 600 mm diameters, connects two parallel shafts 3 m apart. The power to be transmitted is 4 kW from the smaller pulley that rotates at 300 rpm. The coefficient of friction between belt and pulley is 0.3 and the safe working tension is 10 N per mm width. Determine (1) Minimum Width of the belt. (2) Length of the belt require? [5]
6. Discuss different gear trains? What is positive drive and negative drive? [3]
7. Gear 2 rotates at 1200 rpm is counter clockwise direction and engages with Gear 3 and Gear 4 mounted on the same shaft. Gear 5 engages with Gear 4. The numbers of teeth on Gears 2, 3, 4 and 5 are 20, 40, 15 and 30, respectively. The angular speed of Gear 5 is [5]



8. A steam turbine operates under steady flow conditions. It receives steam 7200 kg/hr from the boiler. The steam enters the turbine at enthalpy of 2800 kJ/kg, a velocity of 400 m/min and an elevation of 4 m. The steam leaves the turbine at enthalpy of 2000 kJ/kg, a velocity of 8000 m/min and an elevation of 1 m. Due to radiation, the amount of heat losses from the turbine to the surrounding is 1580 kJ/hr. Calculate the power output of the turbine. [10]
9. A member ABCD is subjected to loading as shown in below Figure. Determine the total elongation. Take $E = 2 \times 10^5 \text{ N/mm}^2$. [5]





NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, South Goa District, Goa-403703

Programme Name: B.Tech (I Semester)
Mid Semester Examinations, October-2025

Roll No

Course Name: Basics of Mechanical and Civil Engineering
Date: 14/10/2025
Duration: 1 hours 30 minutes

Course Code: ME150
Time: 10:00 AM – 11:30 AM
Max. Marks: 50

ANSWER ALL QUESTIONS WITH PROPER DIAGRAM AND SKETCH

- Q1. a. Discuss different modes of Heat Transfers. Which mode of heat transfer is important in our daily life? [3]
- b. The inner surface of a plain brick wall is at 60°C and the outer surface is at 20°C . Calculate the rate of heat transfer per m^2 of the surface area of the wall, which is 260 mm thick. The thermal conductivity of the brick is 0.55 W/mK . [3]
- c. A heat engine having an efficiency of 10 % is used as a refrigerator. And work done on the system is 10 J. How much energy should be taken by the refrigerator from the refrigeration space for enhancing the cooling effect? [5]
- Q2. a. Find the expression of Air-Standard efficiency of Diesel Combustion Cycle. [6]
- b. The minimum pressure and temperature in an Otto cycle are 100 kPa and 27°C respectively. If the compression ratio of the cycle is 8:1 and the amount of heat added to the air per cycle is 1500 kJ/kg. (i) Determine the pressure and temperature at all points of the cycle. (ii) Also calculate the specific work and thermal efficiency. Take for air $C_v = 0.718 \text{ kJ/kg.K}$ and $\gamma = 1.4$. [5]
- Q3. a. Discuss the first law of thermodynamics for control volume. And derive the steady flow energy equation. [6]
- b. A heat engine is supplied with 1200 KW of heat from a constant temperature source of 600 K and it rejects 700 KW of heat to a sink at 300 K. Which of the following is true for this heat engine? If the reversibility of engine is defined by $\eta_{\text{Carnot}} = \eta_{\text{actual}}$ and Irreversibility defined by $\eta_{\text{Carnot}} \geq \eta_{\text{actual}}$ and impossible engine defined by $\eta_{\text{Carnot}} \leq \eta_{\text{actual}}$. [5]
- Q4. a. Discuss various types of stresses and strains. What do you mean by modulus of elasticity and Poisson's ratio? [6]
- b. A bar of 20 mm diameter subjected to a pull of 50 kN. The measured extension on gauge length of 250 mm is 0.12 mm and change in diameter is 0.00375 mm. Calculate: (i) Young's modulus (E) (ii) Poisson's ratio (μ) and (iii) Shear modulus (G). [5]
- Q5. Discuss the stress and strain diagram for mild steel. Why are we considering engineering stress rather than true stress in engineering application? [6]



Roll No	2	5	E	C	E	1	0	3	7
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NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech., II Semester

Mid-Semester Examination, March-2026

Course Name: Basics of Mechanical and Civil Engineering

Date: 9th March., 2026

Duration: 1 Hour 30 Minutes

Course Code: ME150

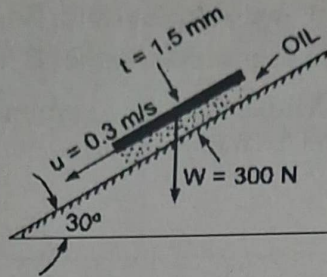
Time:-10:00 AM-11:50 AM

Max. Marks: 50

(ANSWER ALL QUESTIONS)

- Q.1. With the help of neat sketch construction and working mechanism of bourden pressure gauge. [5 Marks]
- Q.2. What are thermodynamic properties? Differentiate between intensive and extensive properties. [3 Marks]
- Q.3. Explain Clausius and Kelvin-Plank statements with neat diagrams. [6 Marks]
- Q.4. State and show that energy is a property of the system [4 Marks]
- Q.5. Calculate the capillary rise in a glass tube of 3 mm diameter when immersed vertically in (a) water and (b) mercury. Take surface tensions $\sigma = 0.0725$ N/m for water and $\sigma = 0.52$ N/m for mercury in contact with air. The specific gravity for mercury is given as 13.6 and angle of contact = 120° . [4 Marks]
- Q.6. A gas undergoes a thermodynamic cycle consisting of the following processes: (i) Process 1-2: Constant pressure $p = 1.4$ bar, $V_1 = 0.028$ m³, $W_{12} = 10.5$ kJ, (ii) Process 2-3: Compression with $pV = \text{constant}$, $U_3 = U_2$, (iii) Process 3-1: Constant volume, $U_1 - U_3 = -26.4$ kJ. There are no significant changes in KE and PE. (a) Sketch the cycle on a p-V diagram. (b) Calculate the net work for the cycle in kJ. (c) Calculate the heat transfer for process 1-2 (d) Show that $\Sigma Q_{\text{cycle}} = \Sigma W_{\text{cycle}}$. [6 Marks]
- Q.7. A nozzle is a device for increasing the velocity of a steadily flowing stream. At the inlet to a certain nozzle, the enthalpy of the fluid passing is 3000 kJ/kg and the velocity is 60 m/s. At the discharge end, the enthalpy is 2762 kJ/kg. The nozzle is horizontal and there is negligible heat loss from it. (a) Find the velocity at exists from the nozzle. (b) If the inlet area is 0.1 m² and the specific volume at inlet is 0.187 m³/kg, find the mass flow rate. (c) If the specific volume at the nozzle exit is 0.498 m³/kg, find the exit area of the nozzle. [6 Marks]
- Q.8. Calculate the dynamic viscosity of an oil, which is used for lubrication between a square plate of size 0.8 m x 0.8 m, and an inclined plane with angle of inclination [4 Marks]

30° as shown in Fig. The weight of the square plate is 300 N and it slides down the inclined plane with a uniform velocity of 0.3 m/s. The thickness of oil film is 1.5 mm.



Q.9. In a S.I. engine working on the ideal Otto cycle, the compression ratio is 6.5. The pressure and temperature at the beginning of compression are 2 bar and 32°C respectively. The peak pressure is 40 bar. Determine the pressure and temperatures at the salient points, the air-standard efficiency and the mean effective pressure. Assume ratio of specific heats to be 1.4 for air. **[8 Marks]**

Q.10. A plane wall is 200 mm thick and its wall area is 6m². If its conductivity is 10.45 W/m°C and surface temperature are steady at 170°C and 60°C, Determine (i) Heat flow across the plane wall (ii) Temperature gradient in the flow direction. **[4 Marks]**



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech. I Year

End Semester Examinations, December-2025

Roll No	2	5	C	5	E	1	0	4	9
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Course Name: Basics of Mechanical and Civil Engineering

Course Code: ME150

Date: 16/12/2025

Time: 10AM to 01PM

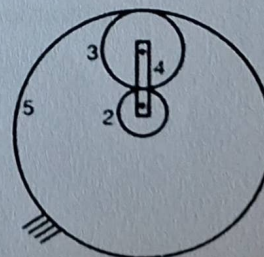
Duration: 3 hours

Max. Marks: 100

ANSWER ALL QUESTIONS WITH PROPER DIAGRAM AND SKETCH

Mechanical Section

1. (a) Discuss the classification of fluid? Give two examples on each from ideal plastic and Newtonian fluid. 4
 - (b) Velocity distribution of fluid flow over a flat plate is given by equation $3 + 0.5y - y^2$ (where y is the vertical distance from the plate surface). Select the correct value of shear stress at a vertical distance of 0.2 m if the dynamic viscosity of the fluid is $0.7 \text{ N} \cdot \text{s}/\text{m}^2$ 4
 - (c) Dimension of surface tension is? 2
 2. (a) What is basic difference between the Open and Closed belt drive? Find the length of Open belt drive? 1.5+2
 - (b) An open belt running over two pulleys 240 mm and 600 mm diameters, connects two parallel shafts 3 m apart. The power to be transmitted is 4 kW from the smaller pulley that rotates at 300 rpm. The coefficient of friction between belt and pulley is 0.3 and the safe working tension is 10 N per mm width. Determine (1) Minimum Width of the belt. (2) Length of the belt require? 4
 3. (a) Discuss different gear trains? What is positive drive and negative drive? 2+1.5
 - (b) An epicyclic gear train is shown schematically in the adjacent figure. The sun gear 2 on the input shaft is a 20 teeth external gear. The planet gear 3 is a 40 teeth external gear. The ring gear 5 is a 100 teeth internal gear. The ring gear 5 is fixed and the gear 2 is rotating at 60 rpm *ccw* (*ccw* = counter clockwise and *cw* = clockwise) 4
- The arm 4 attached to the output shaft will rotate at

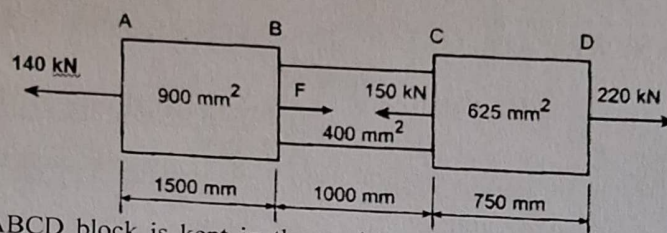


4. A steam turbine operates under steady flow conditions. It receives steam 7200 kg/hr from the boiler. The steam enters the turbine at enthalpy of 2800 kJ/kg, a velocity of 400 m/min and an elevation of 4 m. The steam leaves the turbine at enthalpy of 2000 kJ, a velocity of 8000 m/min and an elevation of 1 m. Due to radiation, the amount of heat losses from the turbine to the surrounding is 1580 kJ/hr. Calculate the power output (kW) of the turbine.

10

5. a A member ABCD is subjected to loading as shown in below figure and the loading is in equilibrium. Determine the total elongation of the ABCD block using the principle of superposition. Take $E = 5 \times 10^5 \text{ N/mm}^2$.

4+1



- b If the same ABCD block is kept in the environment of 100°C and there is **no external** force applied on it then what is the thermal stress developed in the block?
6. (a) What is casting process? Discuss the function of riser? Write four defects of casting. 1+2+2
(b) Write the classification of welding? Thoroughly discuss the Arc welding process. Write the four welding defects. 1+2+2
7. (a) Consider a black body with emissivity, $\epsilon = 1$, a surface area of 400 cm^2 , and a temperature of 200°C . This body is kept in a room at a temperature of 30°C . Using the Stefan Boltzmann law, calculate the net radiation by the body. Take Stefan Boltzmann constant $5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$. 2.5
(b) Air at 30°C is flowing over a flat plate of size $0.5\text{m} \times 0.2\text{m}$. If the convection heat transfer coefficient is $150 \text{ W/m}^2^\circ\text{C}$, determine the heat transfer rate from the air to one side of the plate when the plate is maintained at 130°C . 2.5
8. (a) Explain in details about the first and second law of thermodynamics? 5
(b) A system is taken through a series of processes as a result of which it is restored to the initial state. The work and heat interactions for some of the processes are measured and they are as given below. Fill the blanks space in the table of the energy interaction. 5

Process	W (kJ)	Q (kJ)	U (kJ)
1-2	100	100	--
2-3	--	-150	200
3-4	-250	--	--
4-1	300	--	50

- (c) A frictionless piston-cylinder device contains a gas initially at 0.8MPa and 0.015m³. It expands quasi-statically at constant temperature to a final volume of 0.030m³. The work output (in kJ) during this process will be ? 5
9. (a) What is Bell Coleman Cycle? Describe in details about the use of this cycle and find the efficiency of the cycle with proper P-V and T-S diagram? 5
- (b) An oil engine works in Diesel cycle with temperature of 27°C at the beginning of compression. If the ratio of adiabatic compression is 14 and that of adiabatic expansion is 8, Find the thermal efficiency of the cycle? 5

9.42

$$\gamma = 1.4$$

Civil Section

10. a) List out the classification of surveying based on purpose, instrument, method, place, and accuracy. 5
- b) List out the types of buildings as per NBC 2
- c) List out the types of building Plans 2
- d) List out the processes for setting out of building. 2
- e) List out the types of flooring and roofing. 2
- f) List out the types of plaster and finishing. 2
- g) List out the types of conventional and modern building materials. 2
- h) List out the sources of raw water. 1
- i) List out the components of roof top rainwater harvesting system 2

*****Best of Luck to Everyone*****



Roll No	2	5	E	C	E	1	0	3	7
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NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech., First Year

End Semester Examination, May-2026

Course Name: Basics of Mechanical and Civil Engineering

Date: 7th May 2026

Duration: 3 Hours

Course Code: ME150

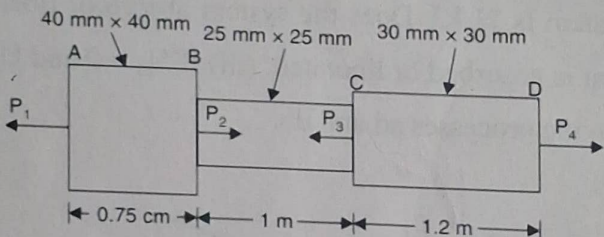
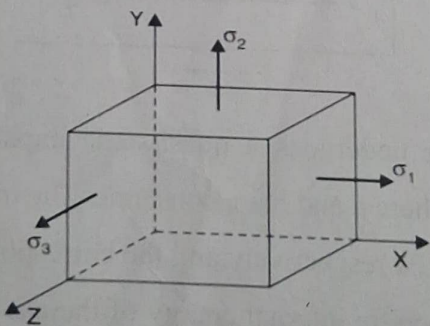
Time: 10:00 A.M. - 1:00 P.M.

Max. Marks: 100

(ALL QUESTIONS ARE COMPULSORY)

PART A: Basics of Mechanical Engineering

Q.1.	(a) Note on concept of Continuum with neat diagram	(3 Marks)
	(b) When a system is taken from state a to state b, in Fig., along path acb, 84 kJ of heat flow into the system, and the system does 32 kJ of work. (i) How much will the heat that flows into the system along path adb, if the work done is 10.5 kJ. (ii) When the system is returned from b to a along the curved path, the work done on the system is 21 kJ. Does the system absorb or liberate heat, and how much of the heat is absorbed or liberated. (iii) If $U_a = 0$ and $U_d = 42$ kJ, find the heat absorbed in the processes ad and db. 	(6 Marks)
	(c) A gas of mass 1.5 kg undergoes a quasi-static expansion which follows a relationship $p=a+bV$, where a and b are constants. The initial and final pressures are 1000 kPa and 200 kPa respectively and the corresponding volumes are 0.20 m ³ and 1.20 m ³ . The specific internal energy of the gas is given by the relation $U= 1.5 pv - 85$ kJ/kg where p is the kPa and v is in m ³ /kg. Calculate the net heat transfer and the maximum internal energy of the gas attained during expansion.	(6 Marks)
Q.2.	(a) With the help of neat sketch explain working mechanism of Refrigeration and Heat pump system. Write down C.O.P. of both systems.	(6 Marks)
	(b) Air flows steadily at the rate of 0.5 kg/s through an air compressor, entering at 7	(6 Marks)

	m/s velocity, 100 kPa pressure, and $0.95 \text{ m}^3/\text{kg}$ volume, and leaving at 5 m/s, 700 kPa, and $0.19 \text{ m}^3/\text{kg}$. The internal energy of the air leaving is 90 kJ/kg greater than that of the air entering. Cooling water in the compressor jackets absorbs heat from the air at the rate of 58 kW. (i) Compute the rate of shaft work input to the air in kW. (ii) Find the ratio of the inlet pipe diameter to outlet pipe diameter.	
Q.3.	(a) Derive equation for thermal efficiency and work output of Otto cycle. (6 Marks) (b) A gas engine operating on the ideal Otto cycle has a compression ratio of 6:1. The pressure and temperature at the commencement of compression are 1 bar and 27°C. Heat added during the constant volume combustion process is 1170 kJ/kg. Determine the peak pressure and temperature, work output per kg of air and air-standard efficiency. Assume $C_v = 0.717 \text{ kJ/kg K}$ and $\gamma = 1.4$ for air. (6 Marks)	
Q.4.	(a) A member ABCD is subjected to point loads P_1, P_2, P_3 and P_4 as shown in figure. Calculate the force P_3 necessary for equilibrium if $P_1 = 120 \text{ kN}$, $P_2 = 220 \text{ kN}$ and $P_4 = 160 \text{ kN}$. Determine also the net change in the length of the member. Take $E = 200 \text{ GN/m}^2$  (b) Write down the total strain for each direction x, y and z for three-dimensional stress system. (5 Marks) 	
Q.5.	(a) Explain Conduction, convection and Radiation. Write down Heat Transfer equations for each of them. (4 Marks) (b) With the help of neat diagram explain working mechanism of Laser machining process. (5 Marks)	
Q.6.	(a) With the help of neat sketch explain working principle of radial drilling machine. Also write its advantages, disadvantages and applications. (5 Marks) (b) With the help neat sketches explain following terms of casting process. (i) Cope (5 Marks)	

	and Drag. (ii) Pattern and Core. (iii) Sprue, Runner and Gate.	
(a)	Derive Ratio of Tensions $\frac{T_1}{T_2} = e^{\mu\theta}$ for a belt drive.	(5 Marks)
(b)	A 100 mm wide and 10 mm thick belt transmits 5 kW of power between two parallel shafts. The distance between the shaft centers is 1.5 m and the diameter of the smaller pulley is 440 mm. The driving and the driven shafts rotate at 60 rpm and 150 rpm respectively. The coefficient of friction is 0.22. Find the stress in the belt if the two pulleys are connected by (i) an open belt and (ii) a cross belt. Take $\mu = 0.22$.	(6 Marks)

PART B: Basics of Civil Engineering

Q.8.	(a) Define Surveying and mention the uses of surveying in civil engineering. (b) Explain/Define the following (with figure, if required): (i) Principles of Surveying. (ii) Geoid, Plane and Geodetic Surveying.	(5 Marks) (2+3 Marks)
Q.9.	(a) In your opinion, what is the role of civil engineer in the current day scenario? List out any five specializations of civil engineering. (b) Specify the standard size of the brick used in construction sector. Define the terms header and stretcher.	(3 Marks) (2 Marks)
Q.10.	(a) What are shallow and deep foundations. (b) Why standardization of a printed document like national building code is required. List any 3 parts of NBC.	(2 Marks) (3 Marks)
